Data Intensive Computing

Exercise-1

Brais, Dinesh, Arash Behaein
{11111111}, {22222222}, {12324255}

Summer 2024

Introduction:

In this Exercise, the mission is to find the most discriminative words in each category by going through Amazon Review Dataset 2014 which contains millions of reviews on 24 product categories by calculating the value of χ^2_{tc} for these words.

In the end we are going to output the top 75 words in each category with highest χ^2_{tc} value in a descending order. To Achieve this goal, we are going to use a stopwords.txt file which will be used to filter out all the irrelevant words such as "a", "an", "with" etc. and we are going to write a MapReduce job for the required calculations. also, mrjob library has been used for implementation of MapReduce job to simplify the process of pre-processing and running the code on cluster. mrjob library, helps efficiently implement the pre-processing steps required for text classification steps, such as tokenization, case folding, stopword filtering, and calculating chi-square values for discriminating terms. Additionally, mrjob facilitates solid integration with Hadoop clusters, ensuring scalability and performance when processing large datasets.

Problem Overview:

The full task involves processing a large text corpus (Amazon Review Dataset 2014) to select discriminative terms for text classification. dataset contains millions of reviews across 24 product categories. The main challenges to deal with is this task are efficient data processing to calculate chi-square values and selecting the top 75 discriminative terms for each category.

to calculate χ^2_{tc} we are going to use the following formula:

$$\chi^2_{tc} = \frac{(N(AD - BC)^2)}{(A + B)(A + C)(B + D)(C + D)}$$

The values of A, B, C, D and N are defined as follows:

- A = Frequency of term in the same category
- B = Frequency of term in different categories
- C = Frequency of non-term in the same category
- D = Frequency of non-term in different categories
- N = Total Number of Documents or category frequency(total)

Approach and Methodology:

Our approach to solve the task contains four steps:

First step contains the pre_processing job which includes Tokenization, case-folding (to make sure all the text is in lower-case format for consistency) and filtering using the "stopwords.txt" file which contains all the irrelevant words to our experiment.

Second step, using the category_counter.py we find the unique categories in the dataset and count the number of times the were occured in the document.

third step: this step contains calculationg the chi_square for different terms inside the document based on the formula and values that was defiend above. the chi_compute.py contains:

- Mapper: Which pre-process each line of input, tokenize the review text, remove unnecessary words based on stopword.txt, and performs stemming. The mapper emits key-value pairs with the category as the key and words as the value.
- Combiner and Reducer: aggregate word counts and calculate chi-square values for each word in each category. Chi-square values are sorted in descending order, and the top 75 terms per category are preserved.

Running the whole code step by step:

- 1. category_Runner.py: calls the category_counter.py job to count category occurrences and print a text file which contains the unique categories in the document.
- 2. mr_runner: calls the chi_computer.py, for calculating chi-square values, and generates output in the specified format (output.txt).

Conclusion:

In conclusion, our approach effectively addresses the task of selecting discriminative terms for text classification using MapReduce and mr-Job. By pre-processing the data, counting category occurrences, and calculating χ^2_{tc} values, we identified terms that exhibit significant differences between categories.